

KNOWLEDGE & UNDERSTANDING – [18 Marks]

Multiple Choice: Write the CAPITAL letter corresponding to the correct answer on the line provided. [8 Marks]

1. Which statement(s) is/are **false** for the graph of $y = -4 \cos \left[\frac{\pi}{11} (x + 7) \right] - 13$?

- | | |
|------|---|
| I. | The amplitude is -4 . |
| II. | The period is 11 . |
| III. | The phase shift is 7 units to the right. |
| IV. | The vertical displacement is 13 units down. |

- A. I and II only B. I and III only C. I, II and III only D. IV only

C

2. The range of $y = \csc(x)$ is

- A. $(-\infty, \infty)$ B. $(-\infty, -1) \cup (1, \infty)$ C. $(-\infty, -1] \cup [1, \infty)$ D. $[0, \infty)$

C

3. The number of solutions to the equation $\sin(4\theta) = \frac{\sqrt{3}}{2}$, where $0 \leq \theta \leq 2\pi$, is

- A. 2 B. 4 C. 6 D. 8

-1 ~~A D~~

4. The period of the function $f(\theta) = -2 \tan(-6\theta + \pi) - 7$ is

- A. $\frac{\pi}{4}$ B. $\frac{\pi}{6}$ C. 6π D. $\frac{\pi}{6}$

B

5. The graph of $f(x) = \tan(x) - 2025$ has a vertical asymptote at

- A. $\frac{21\pi}{4}$ B. $\frac{11\pi}{3}$ C. $\frac{15\pi}{2}$ D. $-\frac{\pi}{6}$

-1 ~~B~~

6. The maximum value occurs on the graph of $f(x) = 2 \cos\left(x + \frac{\pi}{6}\right) - 1$ when x is equal to

- A. $-\frac{\pi}{6}$ B. $-\frac{7\pi}{6}$ C. $\frac{\pi}{6}$ D. $\frac{5\pi}{6}$

A

7. The solution to $\tan(x) = 1$ over the interval $0 \leq x \leq 2\pi$ is

- A. $x = \frac{\pi}{4}, \frac{5\pi}{4}$ B. $x = -\frac{\pi}{4}$ C. $x = 1$ D. $x = \frac{\pi}{4}$

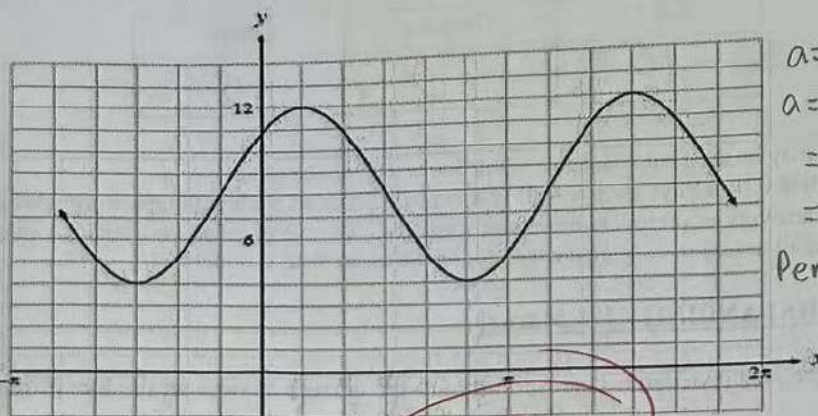
A

8. The y -intercept of the graph represented by $f(\theta) = -3 \sin\left(k\theta + \frac{\pi}{2}\right) - b$ is

- A. $-b$ B. $-(b + 3)$ C. $\frac{3-b}{k}$ D. $\frac{-3-b}{k}$

-1 ~~D~~ B

9. Determine a sine function and a cosine function for the graph below. [4 Marks]



$$a = \frac{\text{max} - \text{min}}{2}$$

$$a = \frac{12 - 4}{2}$$

$$= \frac{8}{2}$$

$$= 4$$

$$\text{Period} = \frac{3\pi}{2} - \frac{\pi}{6}$$

$$= \frac{9\pi}{6} - \frac{\pi}{6}$$

$$= \frac{8\pi}{6}$$

Sine Function: $y = 4 \sin\left[\frac{8\pi}{6}\left(x - \frac{\pi}{6}\right)\right] + 8$

Cosine Function: $y = 4 \cos\left[\frac{8\pi}{6}\left(x + \frac{\pi}{6}\right)\right] + 8$

10. Solve the following. Exact answers. [6 Marks]

a. $4 - 3 \sec^2(x) = 0, x \in [0, 2\pi].$ [3]

b. $2 \sin(3x) - 1 = 0, x \in [0, \pi].$ [3]

$$2 \sin(3x) - 1 = 0$$

$$2 \sin(3x) = 0 + 1$$

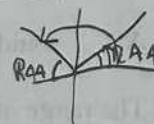
$$2 \sin(3x) = 1$$

$$\sin(3x) = \frac{1}{2}$$

$$\text{Let } A \text{ be } 3x$$

$$\text{RAA} = \sin^{-1}\left(\frac{1}{2}\right)$$

$$\text{RAA} = \frac{\pi}{6}$$



$$A_1 = \frac{\pi}{6}$$

$$3x_1 = \frac{\pi}{6}$$

$$x_1 = \frac{\pi}{6} \div 3$$

$$x_1 = \frac{\pi}{6} - \frac{3}{1}$$

(rest of work on

THINKING - [5 Marks]

11. Anusha is on a roller coaster where her height, in metres, above the ground over time, in seconds, is modelled by the equation $h(t) = a \cos\left[\frac{\pi}{12}(t - 4)\right] + c$. Determine the values of a and c if her height is $2c - 3a + 17$ metres at 16 seconds and her height is $2c + 5a - 133$ metres at 12 seconds. [5 Marks]

$$1. = \frac{4\pi}{3}$$

$$\text{Period} = \frac{2\pi}{|k|}$$

$$\frac{4\pi}{3} = \frac{2\pi}{k}$$

$$4\pi \cdot k = 6\pi$$

$$k = \frac{6\pi}{4\pi}$$

$$k = \frac{3\pi}{2}$$

$$d(\text{sine}) = x_{\max} = \frac{p}{4}$$

$$= \frac{\pi}{6} - \frac{4\pi}{6}$$

$$= \frac{\pi}{6} - \frac{\pi}{3}$$

$$= -\frac{\pi}{6}$$

$$d(\text{cosine}) = \frac{\pi}{6}$$

$$C = \frac{\text{max} + \text{min}}{2}$$

$$= \frac{12 + 4}{2}$$

$$= \frac{16}{2}$$

$$= 8$$

$$10. b) x_1 = \frac{\pi}{6} \times \frac{1}{3}$$

$$= \frac{\pi}{18}$$

$$A_2 = \frac{5\pi}{6}$$

$$3x_2 = \frac{5\pi}{6}$$

$$x_2 = \frac{5\pi}{6} \div 3$$

$$x_2 = \frac{5\pi}{6} \div \frac{3}{1}$$

$$x_2 = \frac{5\pi}{6} \times \frac{1}{3}$$

$$x_2 = \frac{5\pi}{18}$$

$$\frac{17\pi}{18}$$

&

$$\frac{13\pi}{18}$$

Instructions

- Non-graphing calculators may be used but not shared. Notebooks may not be used.
- Only methods taught in MHF4U1 at Bayview S.S. will be accepted. Show all work in the space provided and show all necessary steps. Leave answers exact unless otherwise specified. The use of cellphones, audio or video recording devices, digital music players or email or text-messaging devices during the assessment is prohibited.

APPLICATION - [17 Marks]

1. A Ferris wheel has a radius of 12 metres and completes 1 cycle in 32 seconds. The bottom of the wheel is 3 metres above the ground. Adam and Angela start their ride at the top of the Ferris wheel.
- a. Determine a Cosine function that represents their height above the ground, in metres, as a function of the time, in seconds. [2 Marks]

$$a = \frac{\text{max} - \text{min}}{2}$$

$$a = \frac{15 - 3}{2}$$

$$= \frac{12}{2}$$

$$= 6$$

$$\text{Period} = \frac{2\pi}{|k|}$$

$$32 = \frac{2\pi}{k}$$

$$32k = 2\pi$$

$$k = \frac{2\pi}{32}$$

$$c = \frac{\text{max} + \text{min}}{2}$$

$$c = \frac{15 + 3}{2}$$

$$= \frac{18}{2}$$

$$= 9$$

$$y = 6\sin\left(\frac{\pi}{16}x\right) + 9$$

$$y = 6\cos\left[\frac{\pi}{16}(x - 8)\right] + 9$$

Function: $y = 6\cos\left[\frac{\pi}{16}(x - 8)\right] + 9$

-1.5

- b. After the wheel starts moving, how many seconds will it take for Adam and Angela to be 19 metres above the ground for the 1st time? Round your answer to 2 decimal places. [3 Marks]

-3

- c. What is Adam's and Angela's vertical height above the ground after 50 seconds? Round your answer to 2 decimal places. [2 Marks]

$$y = 6\cos\left[\frac{\pi}{16}(x - 8)\right] + 9$$

$$y = 6\cos\left[\frac{\pi}{16}(50 - 8)\right] + 9$$

$$y = 6\cos\left(\frac{21\pi}{8}\right) + 9$$

$$y = 6(-0.35) + 9$$

$$y = -2.28 + 9$$

$$y = 6.72$$

Adam's and Angela's vertical height above the ground after 50 seconds is 6.72m.

please correct

2. The depth of the water at a harbour is given by $d(t) = A \sin \left[\frac{\pi}{6}(t - B) \right] + 10$, $A > 0, B > 0$; $d(t)$ represent the depth of the water in meters, and t be the number of hours after midnight (12:00 AM). Low tide occurs at 4:00 AM, when water depth is 6 m and high tide occurs at 10:00 AM when the water depth is 14 m. Sketch a properly labelled graph of $d(t)$ for the first 24 hours. What is the value of A and B ? [3 Marks]

$$A = \frac{\text{Max} - \text{min}}{2}$$

$$A = \frac{14 - 6}{2}$$

$$= \frac{8}{2}$$

$$= 4$$

$$B = x_{\text{max}} - \frac{P}{4}$$

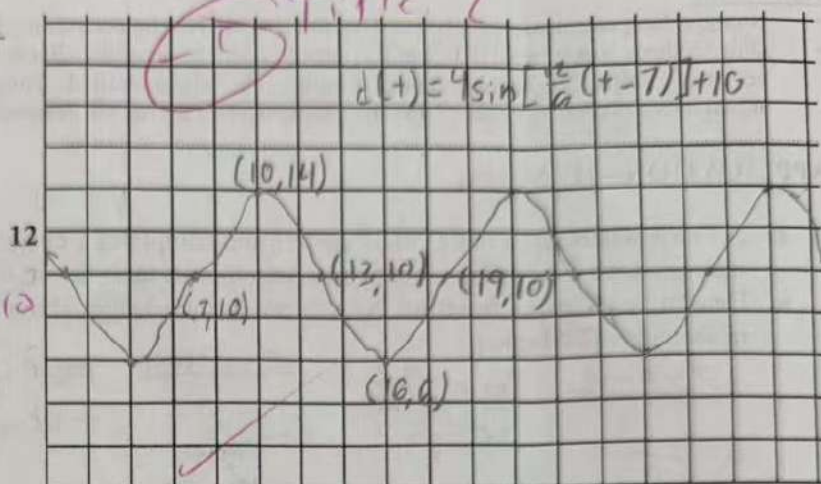
$$\text{Period} = 2 \times (10 - 4)$$

$$= 2 \times 6$$

$$= 12$$

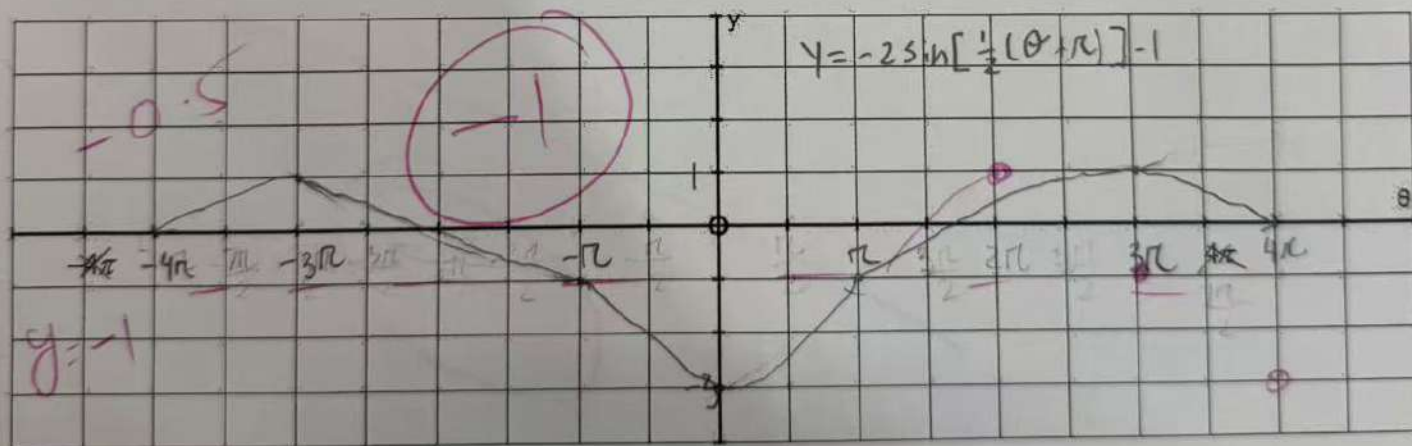
$$B = 10 - \frac{12}{4}$$

$$= 7$$



A: 4 ✓
 B: 7

3. Sketch a properly labelled graph of $y = -2 \sin \left[\frac{1}{2} \theta + \frac{\pi}{2} \right] - 1$ for $-4\pi \leq \theta \leq 4\pi$. [3 Marks]



4. Solve $2 \cos \left(2x - \frac{\pi}{3} \right) - \sqrt{3} = 0$, $x \in [0, 2\pi]$. Exact answer(s). [4 Marks]

MHF4U1 – V1

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THINKING – [5 Marks]

1. Solve $4 \cos(2x) - \sin(x) \csc^3(x) + 2 = 0$, $x \in [0, 2\pi]$. Exact answer(s). [5 Mark]